Biochemistry Cumulative Exam – April 19, 2013

Nobel Laureate John Walker has recently discovered that the F_o subunit of mammalian ATP synthase contains 8 c subunits, 1 a subunit, and two b subunits. The F_1 subunit has been known for many years to contain five subunits with the stoichiometry: $\alpha_3\beta_3\gamma\delta\varepsilon$

A. (30) Discuss the structure of the ATP synthase protein, and how it is oriented in the mitochondrial membrane. Discuss the mechanism by which ATP synthase couples the energy of proton transport to ATP synthesis.

B. (10) Discuss what Dr. Walker's new discovery about the stoichiometry of the c subunits tells us about how many protons need to be transported across the membrane by ATP synthase to synthesize one molecule of ATP.

C. (20) Calculate the exergonic free energy ΔG in kJ/mol for transporting one mole of protons from the outside to the inside of the inner mitochondrial membrane, assuming that $pH_{in} = 8$, $pH_{out} = 7$, $V_{in}-V_{out} = -0.2$ V.

D. (20) Calculate the actual endergonic free energy ΔG in kJ/mol for the synthesis of one mole of ATP by ATP synthase: $ADP + Pi \rightarrow ATP + H_2O$
The standard free energy change $\Delta G^{o\prime}$ for the reaction is +30.5 kJ/mol.
Mitochondrial conditions: $[ATP] = 10^{-2}$ M, $[ADP] = 10^{-3}$ M, $[P_i] = 10^{-2}$ M.

E. (20) Is the exergonic free energy of proton transport through ATP synthase large enough to drive the endergonic free energy of ATP synthesis, given the stoichiometry of proton transport by the ATP synthase determined in part B above? Calculate the efficiency by which the free energy of proton transport is coupled to the synthesis of ATP.

Possibly useful equations:

$\Delta G(X_{out} \rightarrow X_{in}) = RT\ln([X_{in}]/[X_{out}]) + ZF(V_{in} - V_{out}) = 5.70 \log_{10}([X_{in}]/[X_{out}]) + ZF(V_{in} - V_{out})$

$\Delta G = \Delta G^o + RT\ln([products]/[reactants]) = \Delta G^o + 5.70 \log_{10}([products]/[reactants])$

R = 8.3 Joule/degree.mol, F = 96.485 KJoule/volt.mol, Z = charge on molecule